

Chapter 3

Intensity Transformation and Spatial Filtering

Part1: Intensity Transformation

3.1 Background

- 3.1.1 Intensity Transform
 - $g[x,y]=T[f[x,y]]$, T is a unary function
- 3.1.2 Spatial Filtering

$$g(x, y) = \sum_{m=-M}^M \sum_{n=-N}^N f(x+m, y+n) \cdot \text{mask}(m, n)$$

3.1.2 Spatial Filtering

*						
	5 *0	5 *-1	4 *0			
	4 *-1	7 *5	4 *-1			
	4 *0	4 *-1	3 *0			
						#

0	-1	0
-1	5	-1
0	-1	0

3.1.2 Spatial Filtering

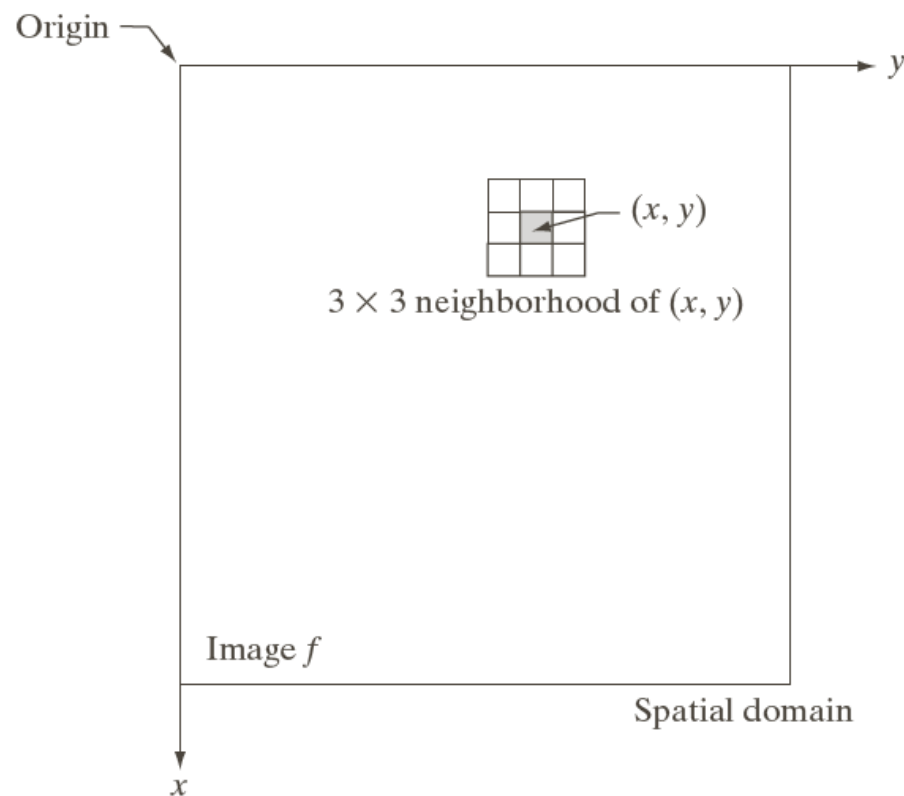


FIGURE 3.1

A 3×3 neighborhood about a point (x, y) in an image in the spatial domain. The neighborhood is moved from pixel to pixel in the image to generate an output image.

Image Enhancement

- Enhancement is the process of manipulating an image so that the result is more suitable than the original for specific application.

3.2 Some Basic Intensity Transform Functions

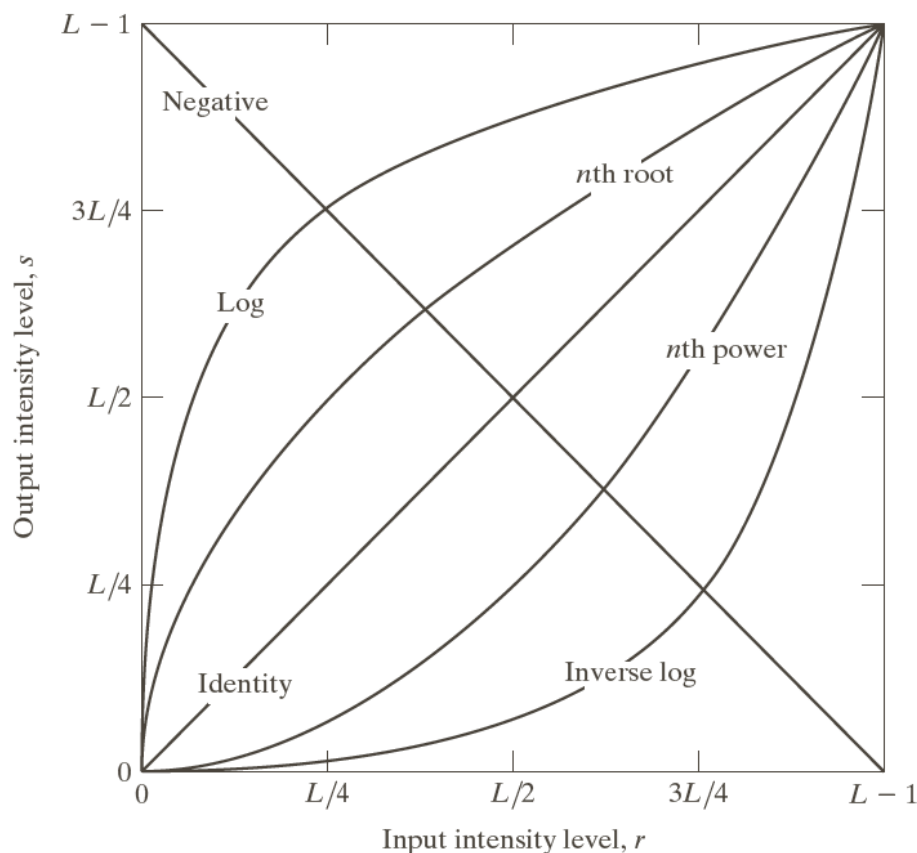
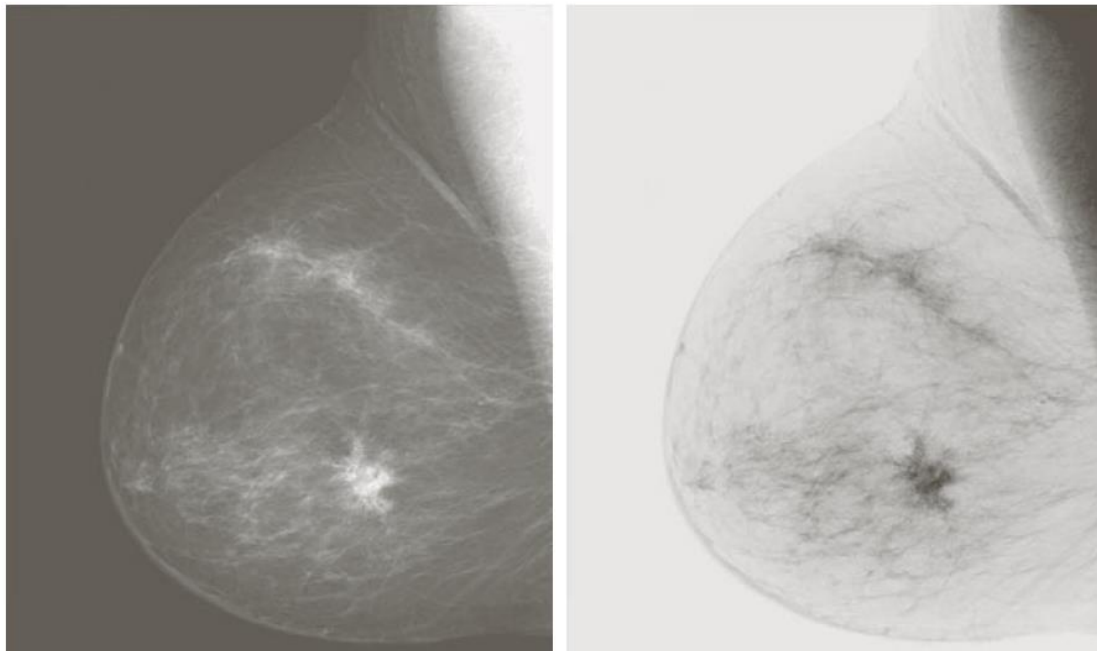


FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.

3.2 Some Basic Intensity Transform Functions

- 3.2.1 Image Negative

$$s = L - 1 - r$$



a b

FIGURE 3.4

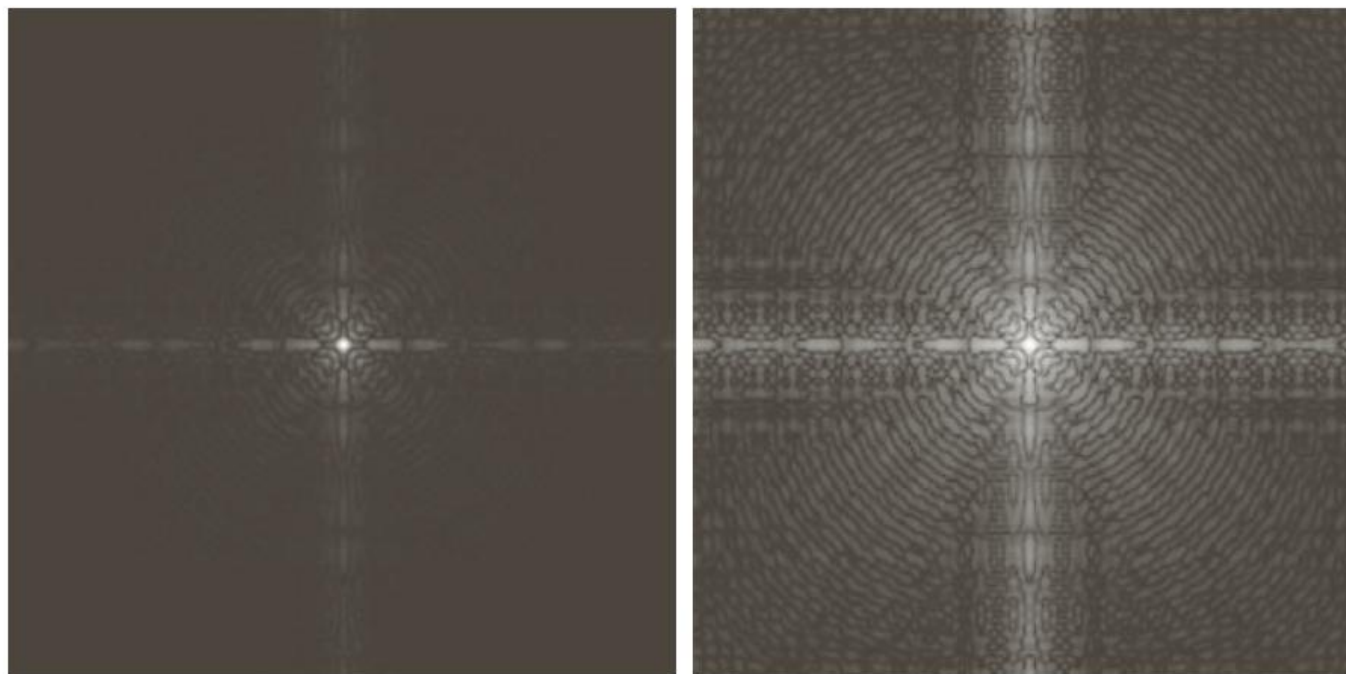
(a) Original digital mammogram.

(b) Negative image obtained using the negative transformation in Eq. (3.2-1).

(Courtesy of G.E. Medical Systems.)

3.2.2 Log Transformations

$$s = c \log(1+r), r \geq 0$$



a b

FIGURE 3.5

(a) Fourier spectrum.
(b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.

3.2.2 Power (Gamma) Transformations

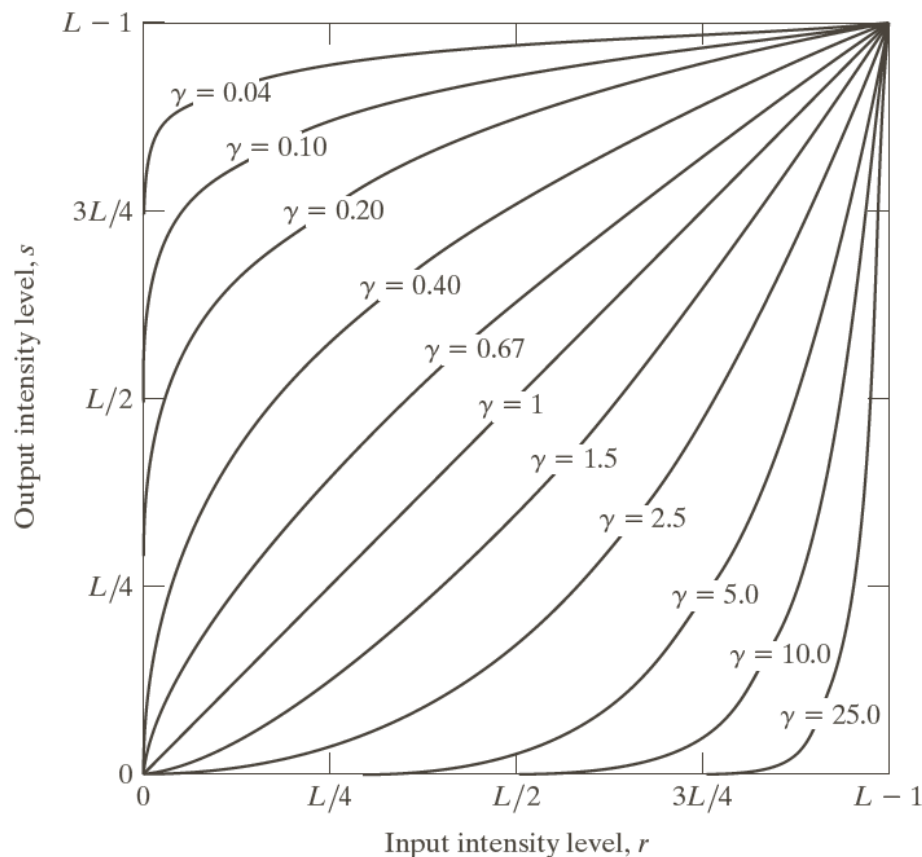
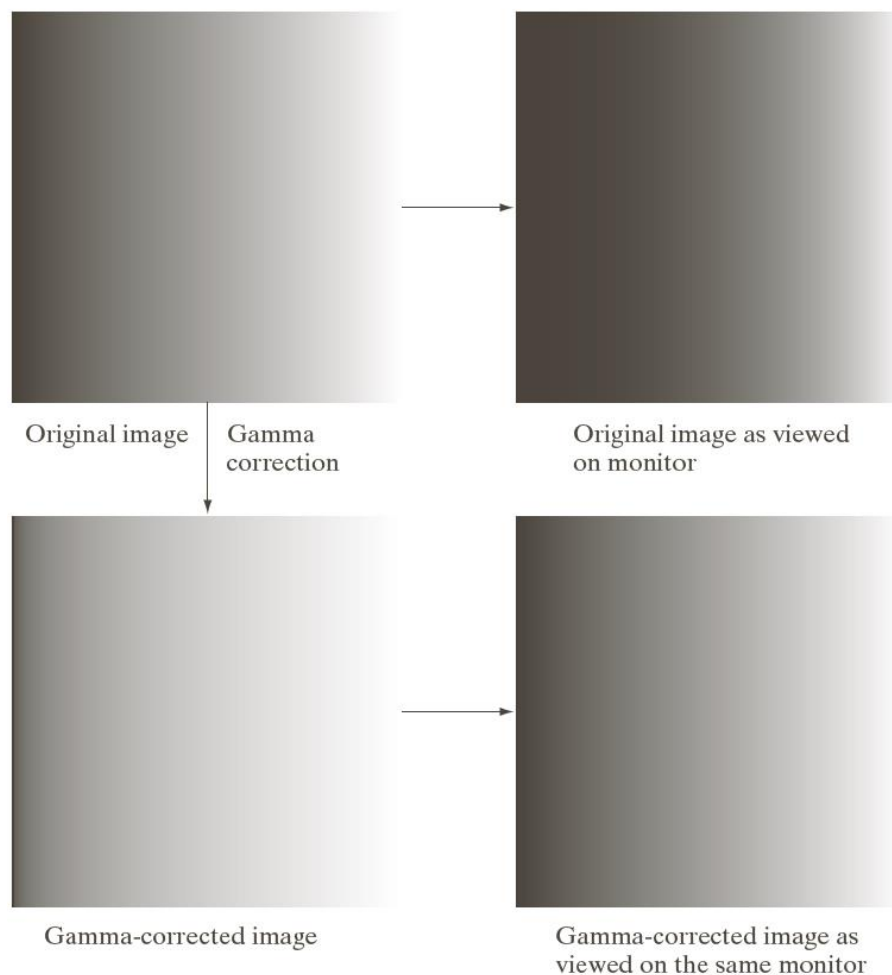


FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.

3.2.2 Power (Gamma) Transformations

- [Programming]
- How to accelerate?

3.2.2 Power (Gamma) Transformations

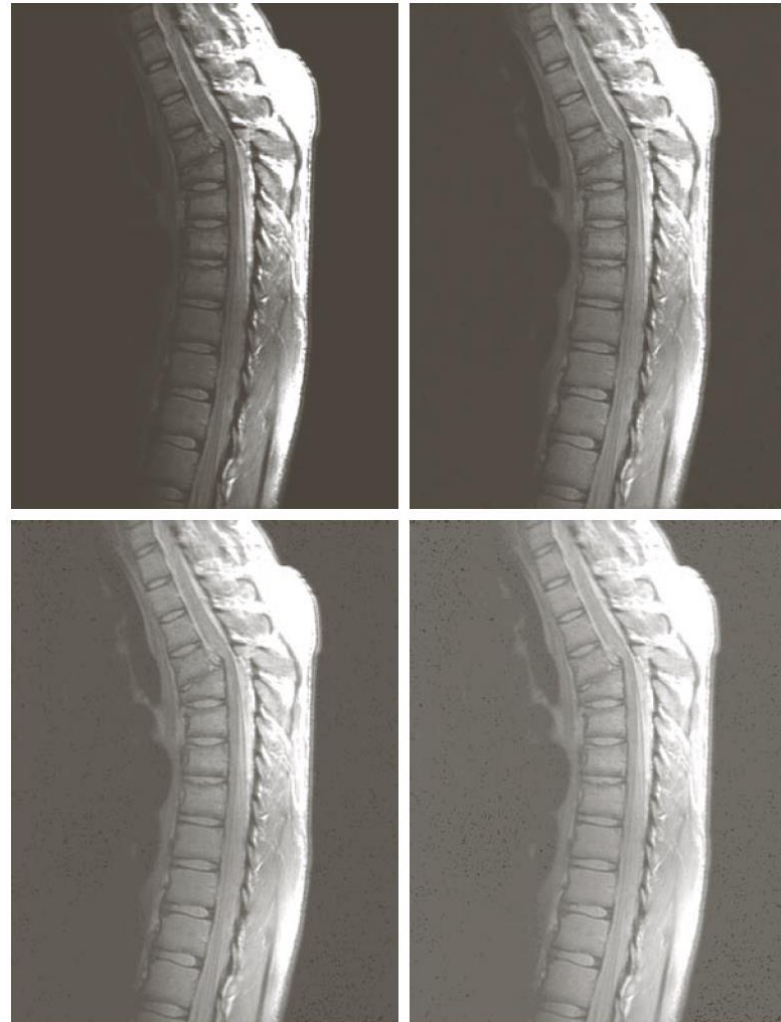


a	b
c	d

FIGURE 3.7

(a) Intensity ramp image. (b) Image as viewed on a simulated monitor with a gamma of 2.5. (c) Gamma-corrected image. (d) Corrected image as viewed on the same monitor. Compare (d) and (a).

3.2.2 Power (Gamma) Transformations



a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine.

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

3.2.3 Power (Gamma) Transformations

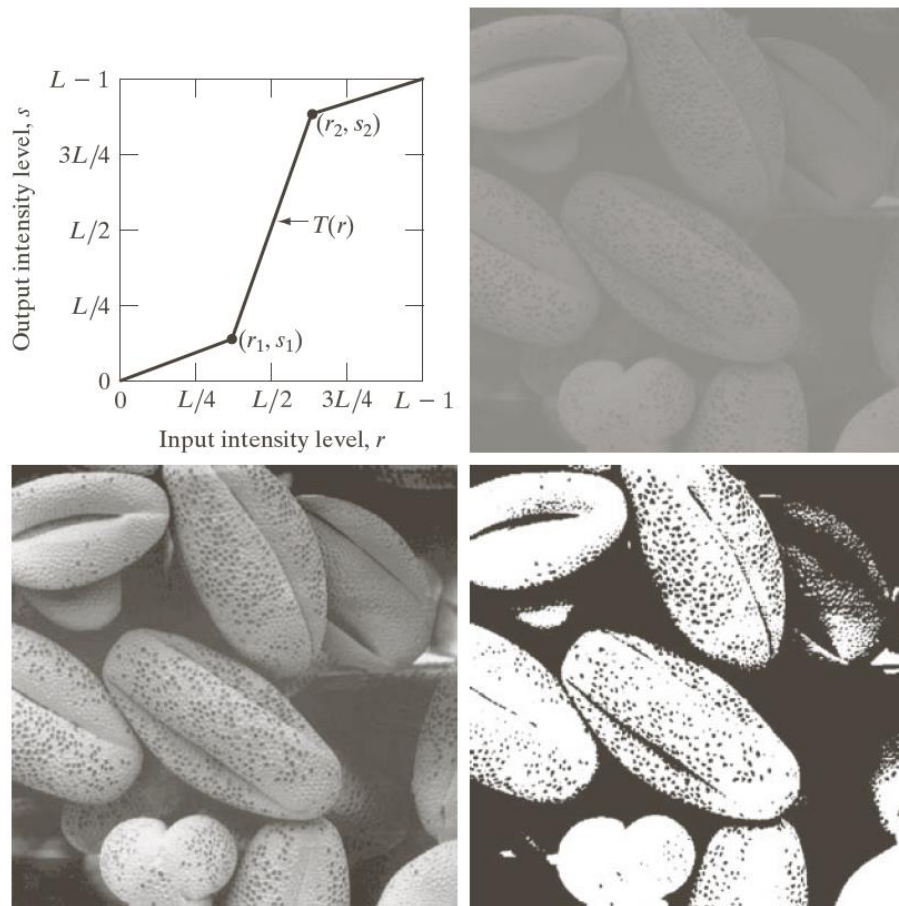


a	b
c	d

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0$, and 5.0 , respectively. (Original image for this example courtesy of NASA.)

3.2.4 Piecewise-Linear Transformation Functions I - Contrast stretching



a	b
c	d

FIGURE 3.10

Contrast stretching.

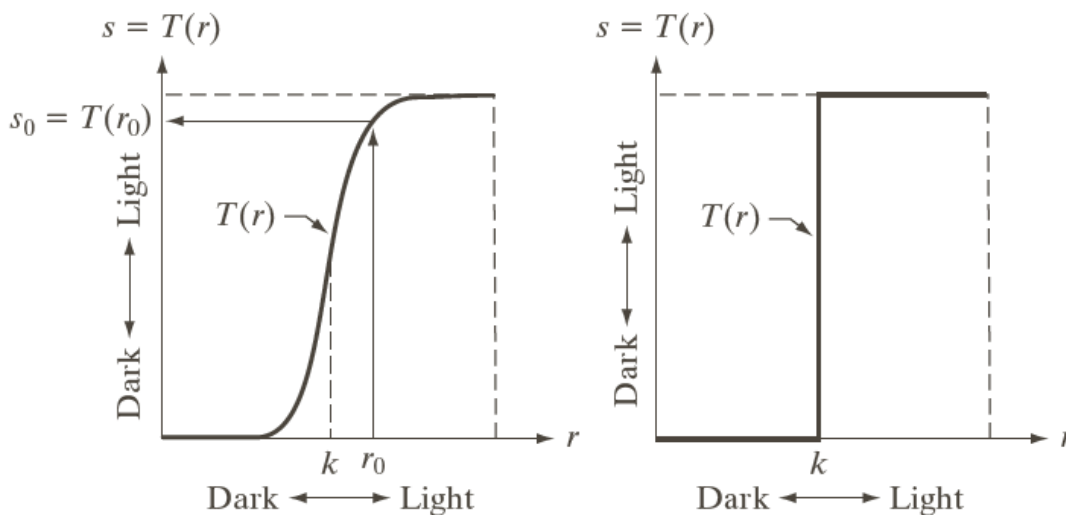
(a) Form of transformation function. (b) A low-contrast image.

(c) Result of contrast stretching.

(d) Result of thresholding.

(Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Thresholding



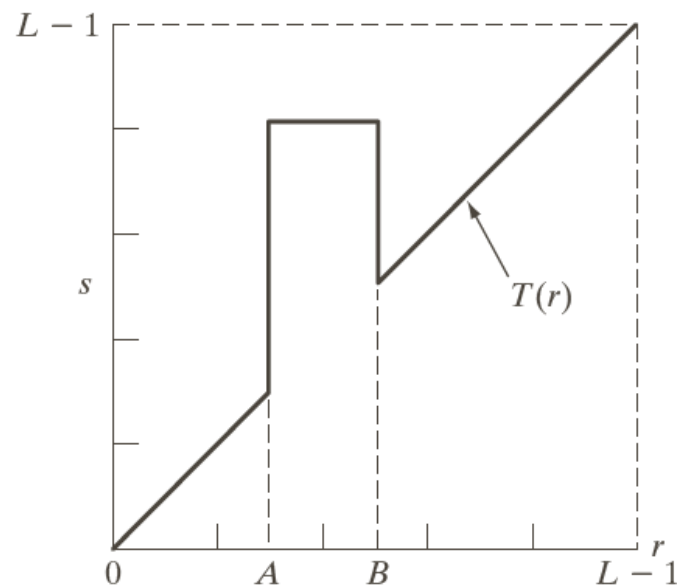
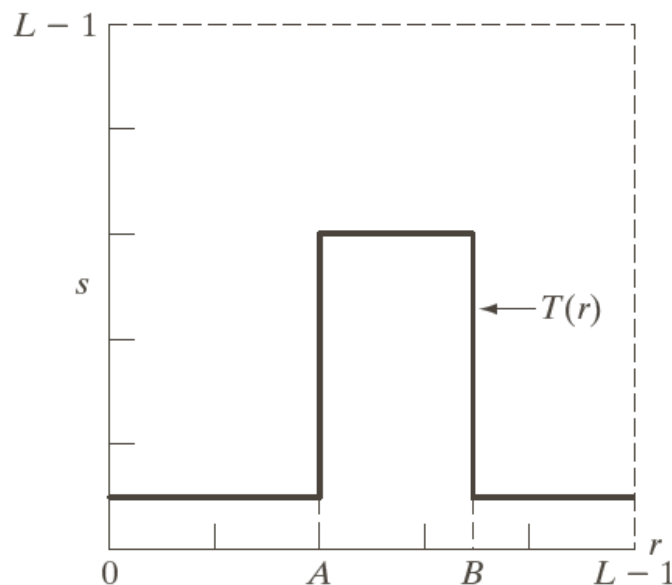
a b

FIGURE 3.2
Intensity transformation functions.
(a) Contrast-stretching function.
(b) Thresholding function.

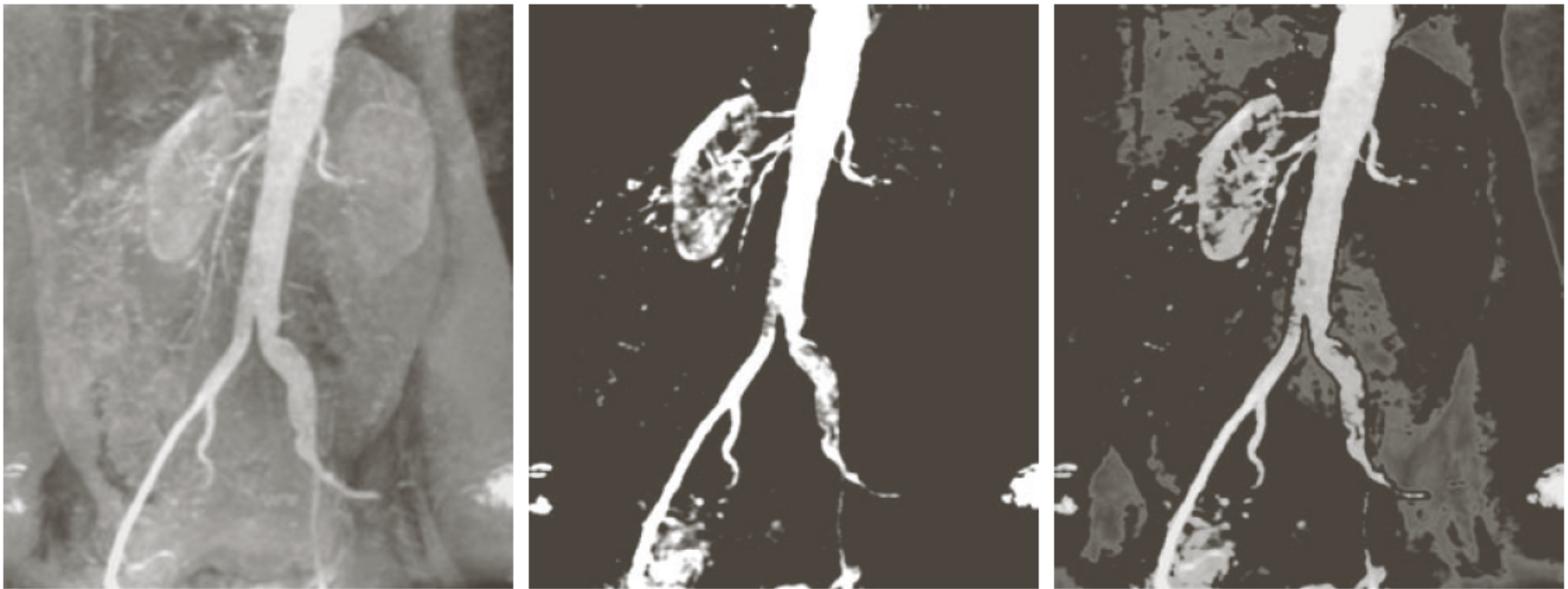
3.2.4 Piecewise-Linear Transformation Functions II: Intensity-level slicing

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.



3.2.4 Piecewise-Linear Transformation Functions II: Intensity-level slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

3.2.4 Piecewise-Linear Transformation

Functions III: Bit-plane slicing

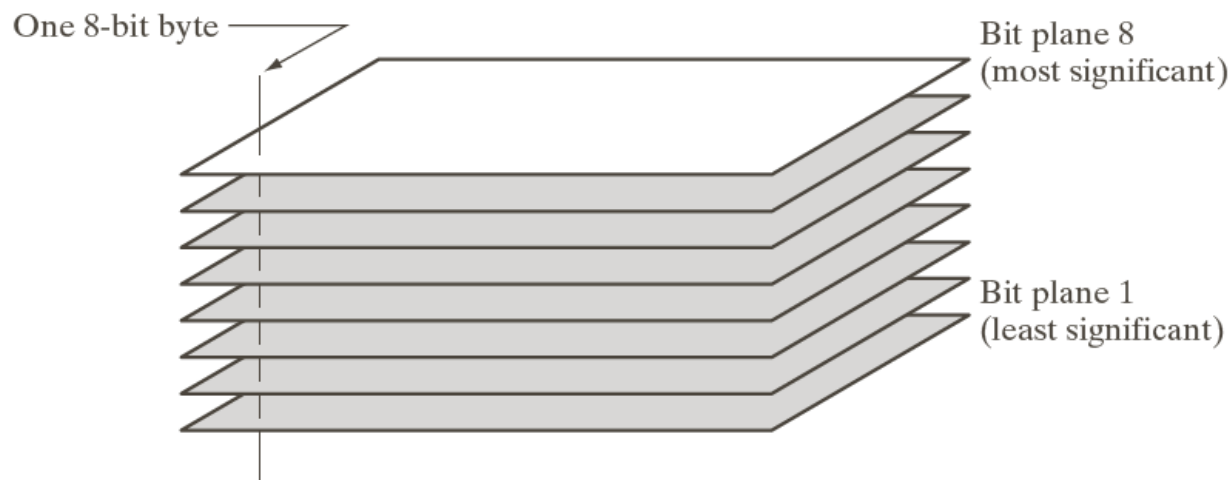


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.

3.2.4 Piecewise-Linear Transformation Functions II: Bit-plane slicing



a	b	c
d	e	f
g	h	i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 500×1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.



Bit-plane slicing

- [Programming]

Bit-plane slicing

- [Draw the bit-plane slicing function curve]

3.2.4 Piecewise-Linear Transformation

Functions II: Bit-plane slicing



a b c

FIGURE 3.15 Images reconstructed using (a) bit planes 8 and 7; (b) bit planes 8, 7, and 6; and (c) bit planes 8, 7, 6, and 5. Compare (c) with Fig. 3.14(a).

3.2.4 Brightness Transformation

Basic

$$g[x,y] = f[x,y] + c$$

$$g[x,y] = k f[x,y]$$

Saturation Operation

$$g[x,y] = S(f[x,y] + c)$$

$$g[x,y] = S(k f[x,y])$$

[Graph & Programming]